

THE SIMPLEX METHOD IN MATRIX FORM

EXAMPLE

$$\begin{aligned} & \text{maximize} && 4x_1 + 3x_2 \\ & \text{subject to} && x_1 - x_2 \leq 1 \\ & && 2x_1 - x_2 \leq 3 \\ & && x_2 \leq 5 \\ & && x_1, x_2 \geq 0. \end{aligned}$$

Form the initial dictionary:

$$\begin{aligned} \zeta &= 4x_1 + 3x_2 \\ x_1 - x_2 + w_1 &= 1 \\ 2x_1 - x_2 + w_2 &= 3 \\ x_2 + w_3 &= 5 \end{aligned}$$

The initial basic indices are $\mathcal{B} = (3, 4, 5)$, the initial nonbasic indices are $\mathcal{N} = (1, 2)$.
The coefficient matrices are:

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & -1 \\ 2 & -1 \\ 0 & 1 \end{bmatrix}.$$

The primal basic entries are: $x_{\mathcal{B}}^* = \begin{bmatrix} x_3^* \\ x_4^* \\ x_5^* \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$. The dual basic entries are: $z_{\mathcal{N}}^* = \begin{bmatrix} z_1^* \\ z_2^* \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$.

Step 1

Primal side

New basic index: $j = 1$

$$\text{Step direction: } \Delta = B^{-1}N \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\text{Step size: } t = \max^{-1} \left\{ \frac{\Delta_3}{x_3^*} = \frac{1}{1}, \frac{\Delta_4}{x_4^*} = \frac{2}{3}, \frac{\Delta_5}{x_5^*} = \frac{0}{5} \right\} = 1$$

New nonbasic index: $i = 3$

$$x_{\mathcal{B}}^* - t\Delta = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix}$$

$$\text{Updated basic vector: } x_{\mathcal{B}}^* = \begin{bmatrix} x_1^* \\ x_4^* \\ x_5^* \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

Updated index lists: $\mathcal{B} = (1, 4, 5)$, $\mathcal{N} = (3, 2)$

$$\text{Updated matrices: } B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & -1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}$$

Dual side

$$z_{\mathcal{N}}^* = \begin{bmatrix} -4 \\ -3 \end{bmatrix} \text{ has negative entries}$$

$$\text{Step direction: } \Delta = -N^{\top}B^{-\top} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\text{Step size: } s = \frac{z_1^*}{\Delta_1} = \frac{-4}{-1} = 4$$

$$z_{\mathcal{N}}^* - s\Delta = \begin{bmatrix} -4 \\ -3 \end{bmatrix} - 4 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -7 \end{bmatrix}$$

$$\text{Updated basic vector: } z_{\mathcal{N}}^* = \begin{bmatrix} z_3^* \\ z_2^* \end{bmatrix} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$$

Step 2

Primal side

New basic index: $j = 2$

$$\text{Step direction: } \Delta = B^{-1}N \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Step size: } t = \max^{-1} \left\{ \frac{\Delta_1}{x_1^*} = \frac{-1}{1}, \frac{\Delta_4}{x_4^*} = \frac{1}{1}, \frac{\Delta_5}{x_5^*} = \frac{1}{5} \right\} = 1$$

New nonbasic index: $i = 4$

$$x_{\mathcal{B}}^* - t\Delta = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

$$\text{Updated basic vector: } x_{\mathcal{B}}^* = \begin{bmatrix} x_1^* \\ x_2^* \\ x_5^* \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

Updated index lists: $\mathcal{B} = (1, 2, 5), \mathcal{N} = (3, 4)$

$$\text{Updated matrices: } B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Step 3

Primal side

New basic index: $j = 3$

$$\text{Step direction: } \Delta = B^{-1}N \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}$$

$$\text{Step size: } t = \max^{-1} \left\{ \frac{\Delta_1}{x_1^*} = \frac{-1}{2}, \frac{\Delta_2}{x_2^*} = \frac{-2}{1}, \frac{\Delta_5}{x_5^*} = \frac{2}{4} \right\} = 2$$

New nonbasic index: $i = 5$

$$x_{\mathcal{B}}^* - t\Delta = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$$

$$\text{Updated basic vector: } x_{\mathcal{B}}^* = \begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix}$$

Updated index lists: $\mathcal{B} = (1, 2, 3), \mathcal{N} = (4, 5)$

$$\zeta_{\max} = 4x_1^* + 3x_2^* = 4 \times 4 + 3 \times 5 = 31$$

Dual side

$$z_{\mathcal{N}}^* = \begin{bmatrix} 4 \\ -7 \end{bmatrix} \text{ has negative entries}$$

$$\text{Step direction: } \Delta = -N^{\top}B^{-\top} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\text{Step size: } s = \frac{z_2^*}{\Delta_2} = \frac{-7}{-1} = 7$$

$$z_{\mathcal{N}}^* - s\Delta = \begin{bmatrix} 4 \\ -7 \end{bmatrix} - 7 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -10 \\ 0 \end{bmatrix}$$

$$\text{Updated basic vector: } z_{\mathcal{N}}^* = \begin{bmatrix} z_3^* \\ z_4^* \end{bmatrix} = \begin{bmatrix} -10 \\ 7 \end{bmatrix}$$

Dual side

$$z_{\mathcal{N}}^* = \begin{bmatrix} -10 \\ 7 \end{bmatrix} \text{ has negative entries}$$

$$\text{Step direction: } \Delta = -N^{\top}B^{-\top} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\text{Step size: } s = \frac{z_3^*}{\Delta_3} = \frac{-10}{-2} = 5$$

$$z_{\mathcal{N}}^* - s\Delta = \begin{bmatrix} -10 \\ 7 \end{bmatrix} - 5 \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\text{Updated basic vector: } z_{\mathcal{N}}^* = \begin{bmatrix} z_4^* \\ z_5^* \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$z_{\mathcal{N}}^* \geq 0$$